GAME ENGINE DESIGN

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How My Game Engine Works

Singleton : Manages a single instance of core managers

GraphicsManager: Handles rendering frames.

ObjectManager: Manages all game objects

-**Object Class**: Represents an individual entity in the game. Position, velocity, and a **Texture**.

-**Texture Class:** Provides visual representation for objects.

PhysicsManager: Handles movement and collisions.

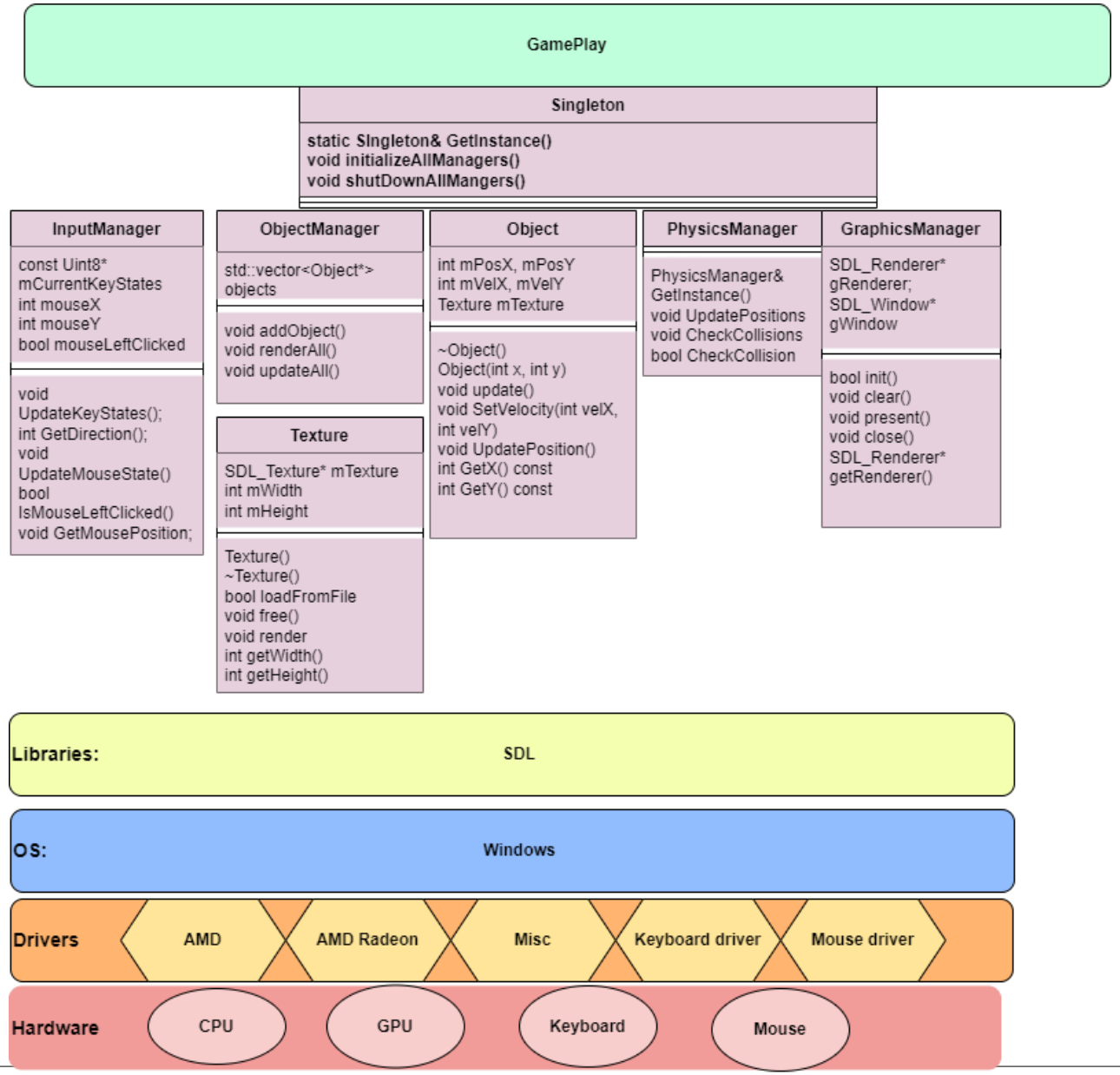
InputManager: Captures player input (keyboard, mouse).

Game Engine

- **Game Play**
- **Singleton:** Initialize and shuts down managers.
- **Managers:** InputManager, ObjectManager, PhysicsManager, GraphicManager

+ **Classes** Object and Texture

- **Libraries:** SDL
- **OS:** Windows
- **Drivers:** AMD, AMD Radeon, Misc, Keyboard + Mouse
- **HW:** CPU, GPU, Keyboard and Mouse



Update Order

•**InputManager**

Captures player inputs.

•**PhysicsManager**:

Updates **object** positions and handles collisions.

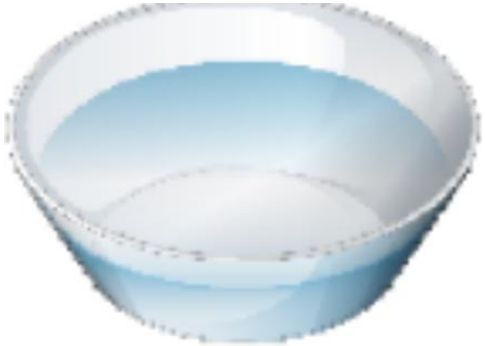
•**ObjectManager**:

Updates and manages all objects. Using the Object Class

•**GraphicsManager**:

Clears the screen and renders objects.

Game Demonstration



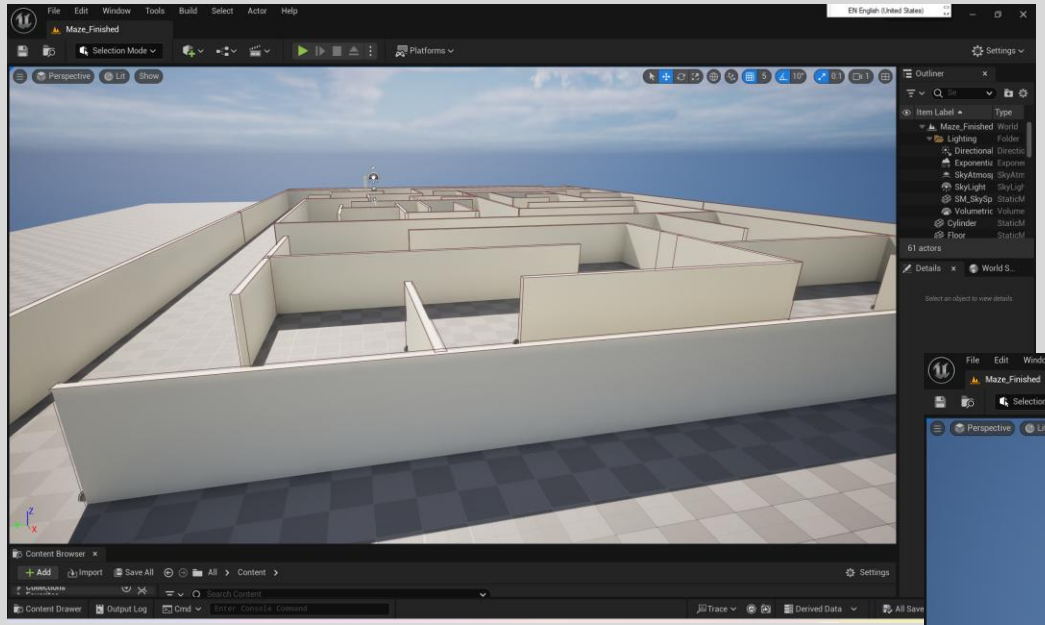
Game 1: Catch My Tears

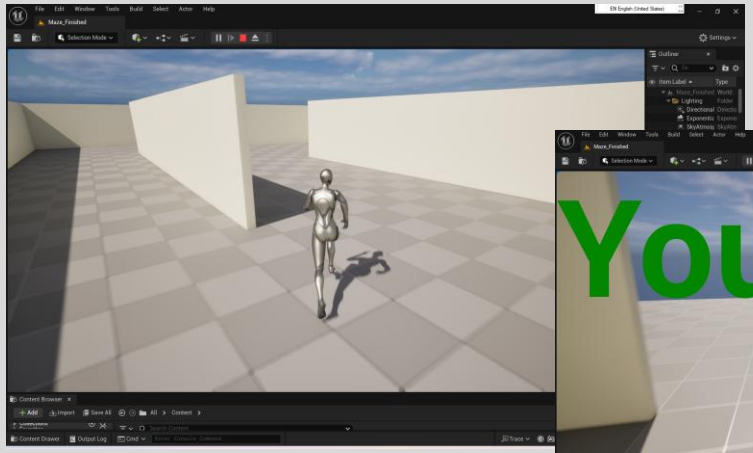
- Move the vase to catch tears while avoiding misses.

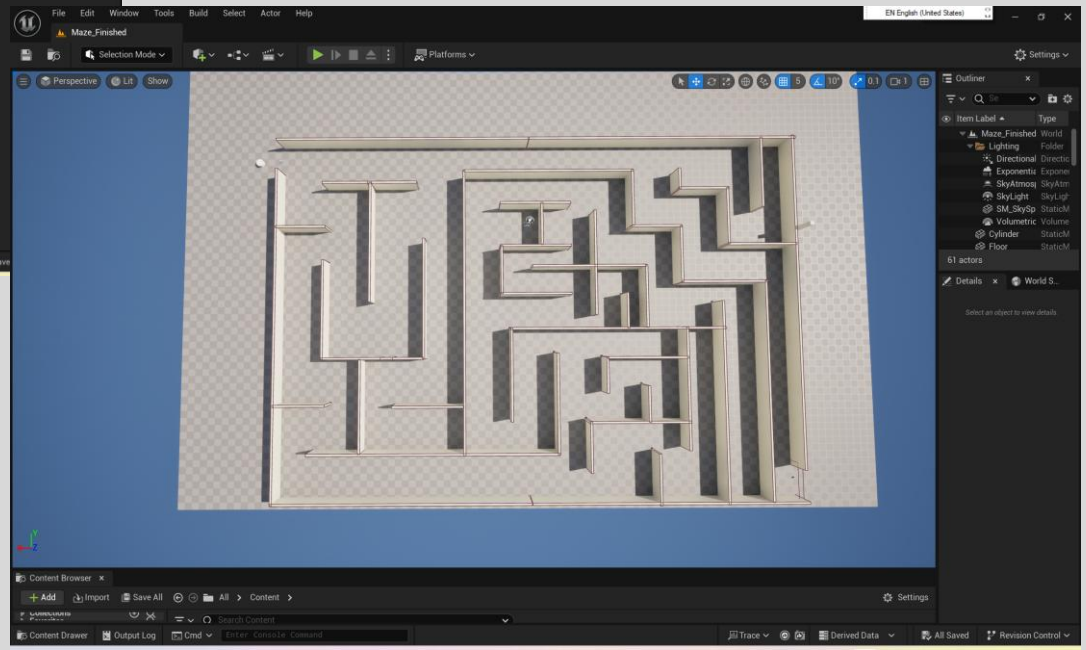
Game 2: Kill the BedBugs

- Click on bedbugs to eliminate them before the screen fills up.

Unreal Engine



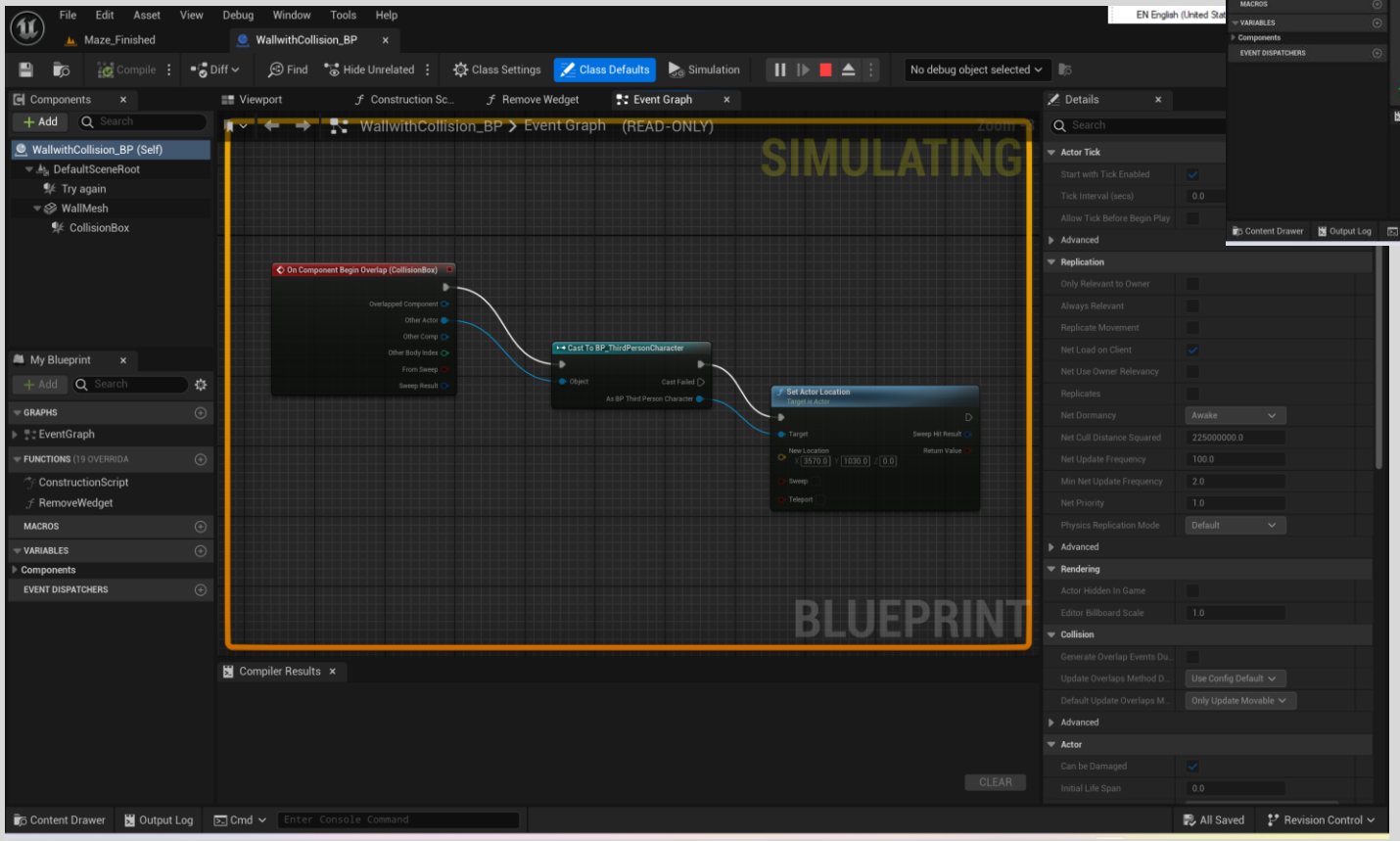


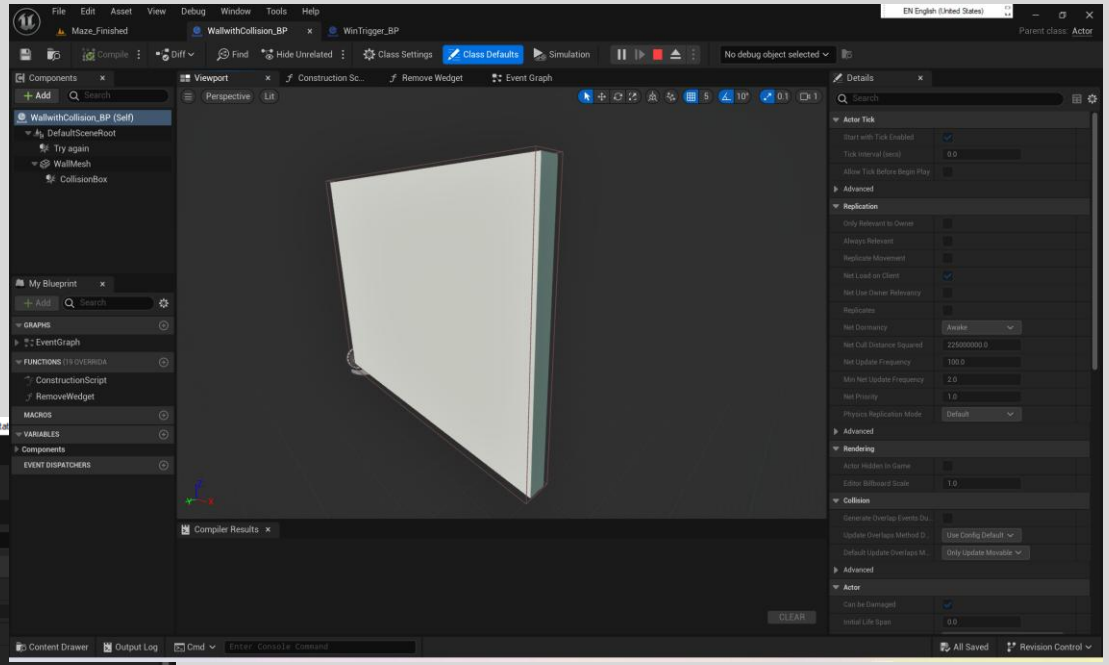


- THE MAZE

Unreal Engine – Blue Prints

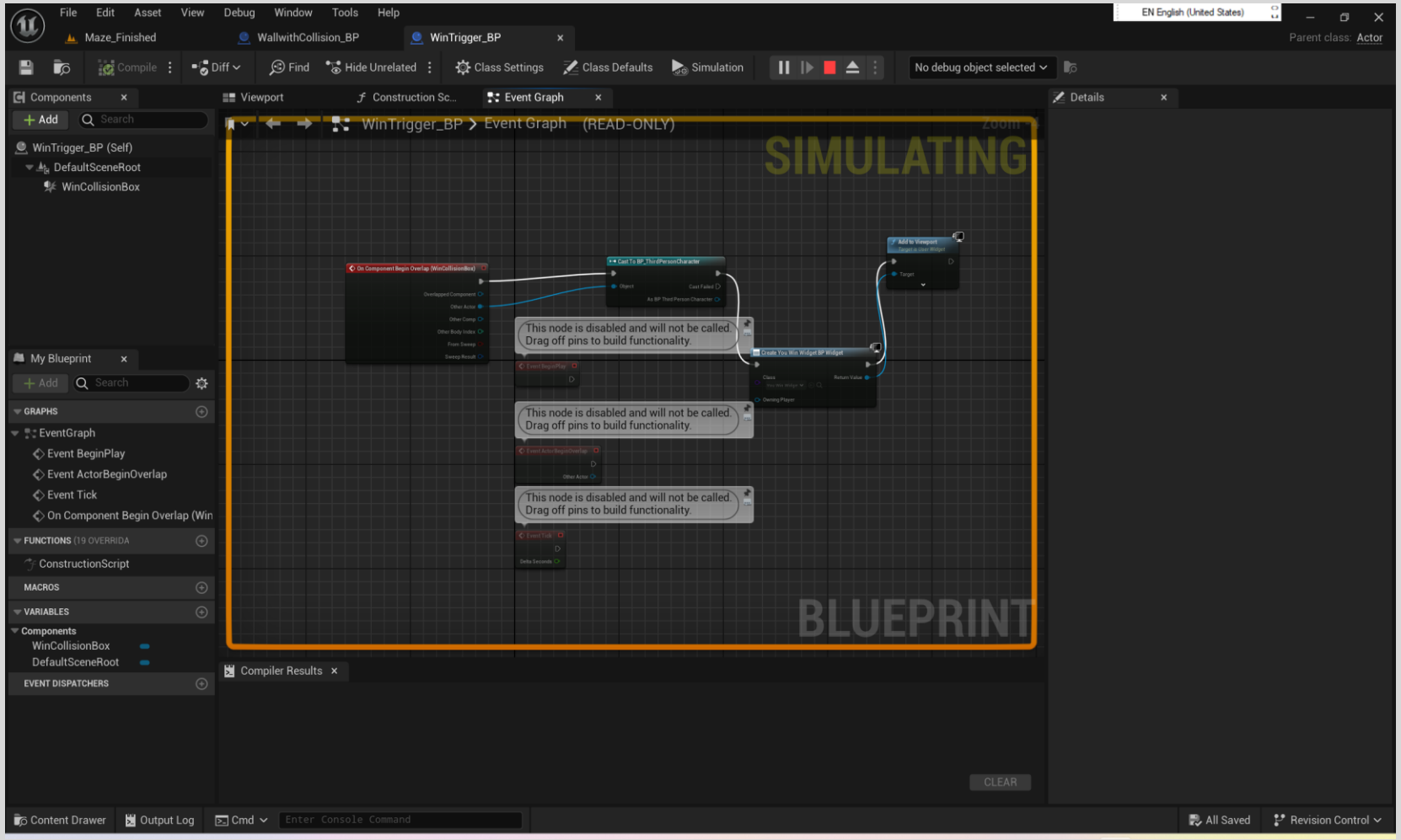
Wall with respawn function





Unreal Engine – Blue Prints

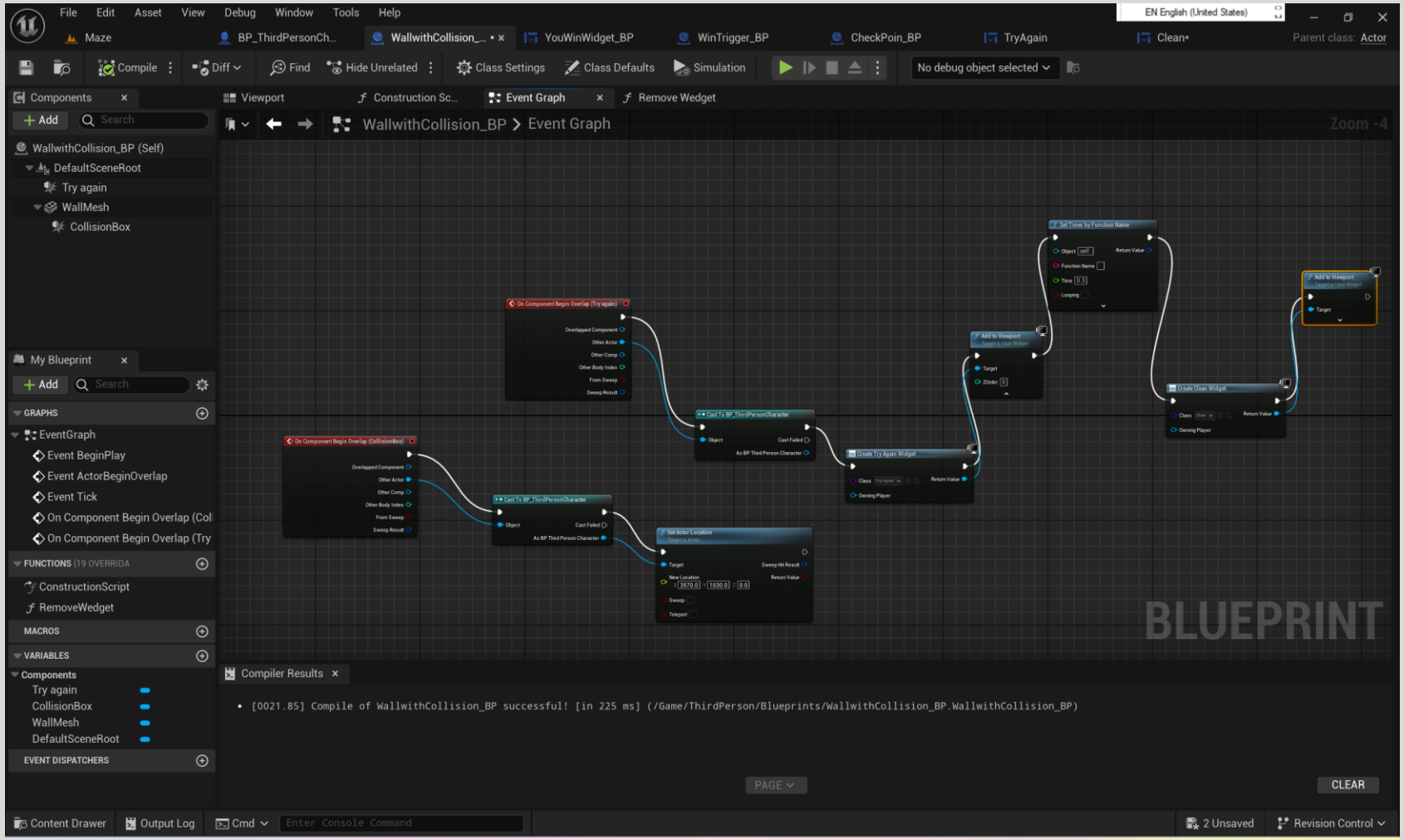
Collision Box with Widget Text



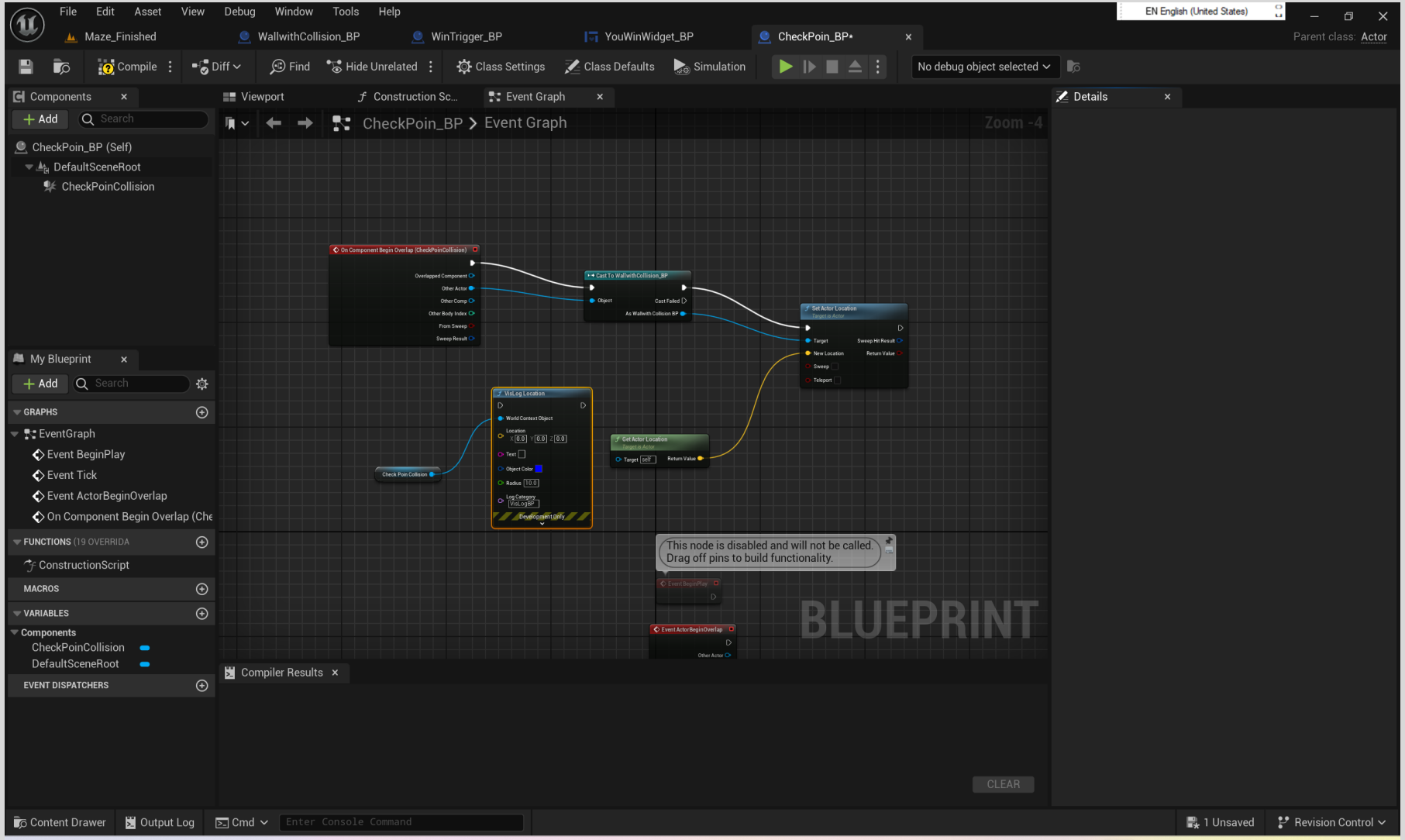
Unreal Engine – Blue Prints

Combination of Both

-unsuccessful



Unreal Engine – Blue Prints

Check Point

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